CET257 Enterprise Project — Group Meeting Minutes: Week 2

Faculty: Faculty of Technology, School of Computer Science
Module: CET257 Enterprise Project
Module Leader: Dr Becky Allen
Supervisory Tutor: Rupak Chaudhary
Team Number: 9 | **Team Name:** SARAS
Meeting Reference: MIN-WK02-20260729 & MIN-WK02-20260731
Dates & Times:

- **Primary Sprint Session:** Wednesday, 29 July 2026 (07:22:47 – 08:08:09 NPT; 45m 22s)
- **Technical Design Walkthrough:** Friday, 31 July 2026 (14:00:00 – 15:15:30 NPT; 75m 30s)
  Location / Platform: Microsoft Teams (Meetings Channel)
  Attendees: Rijan Kapur Poudel (PM/Backend), Sushank Jung Rawal (Deputy PM/UI Lead)

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1. Attendees and Day's Activities

Attendance Register

- **Rijan Kapur Poudel** — Present for both sessions (Full attendance recorded in Teams workbook).
- **Sushank Jung Rawal** — Present for both sessions (Full attendance recorded in Teams workbook).

Day's Activities and Agenda

1. **Architecture & Schema Review:** Evaluating the initial Supabase PostgreSQL database schema (Migration 001) and Next.js full-stack directory architecture.
2. **Frontend UI Walkthrough:** Live inspection of the newly committed lesson wizard (`lesson-experience.tsx`), navigation controls, and card transitions.
3. **Design System & Corporate Identity Review:** Formal review of the SARAS crane logo, typography hierarchy, and Tailwind CSS design tokens.
4. **PACT Analysis Synthesis:** Aligning user stories with the physical context of warehouse logistics operatives (noisy environment, varied device screens, high fatigue).
5. **Project Plan Baseline Review (Task A5):** Formulating the Gantt schedule and Work Breakdown Structure (WBS) across Weeks 1 to 7.
6. **Week 3 Consultation Preparation:** Outlining initial themes for the upcoming formal client consultation with Mr. Ninson Lama.

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2. Breakdown of Achieved Activity for the Week

Rijan Kapur Poudel

- Provisioned Supabase cloud project instance; authored and executed SQL Migration 001 creating core relational tables (`profiles` , `modules` , `lessons` , `user_progress`).

- Configured Supabase email/password authentication and session cookie management in Next.js middleware.
- Authored initial draft of `01 Requirements Specification.pdf` (Task A1), defining 10 functional requirements (FR-01 to FR-10) and 8 non-functional requirements (NFR-01 to NFR-08).
- Built Task A5 Master Project Schedule baseline in Microsoft Excel (`Project Plan.xlsx`), mapping WBS work packages and dependencies.
- Integrated Microsoft Teams webhook notifications for GitHub commit and pull request tracking.

Sushank Jung Rawal

- Built the core responsive layout framework using Tailwind CSS, establishing semantic color classes (`bg-saras-blue` , `text-saras-blue` mapped to `#0056D2`).
- Implemented the interactive step-by-step lesson component (`lesson-experience.tsx` , Commit `ca60157`), enabling smooth forward and backward card navigation.
- Designed the official SARAS crane brand mark in vector SVG and PNG formats, paired with geometric typography for the corporate identity guide.
- Authored `02 User Requirements Analysis.pdf` (Task A1), conducting detailed PACT analysis and heuristic evaluations focused on warehouse safety training.
- Designed visual wireframes for the promotional marketing landing page and course navigation dashboard.

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3. Breakdown of Planned Activity for the Week Ahead (Week 3)

Rijan Kapur Poudel

- Author and dispatch the formal invitation to client Mr. Ninson Lama for the Week 3 consultation (targeted for Thursday, 06 August 2026 at 14:00 NPT).
- Formulate the five focused inquiry questions for the client session covering learners, hazards, devices, feedback, and training boundaries.
- Address database concurrency issues: refactor the `user_progress` table with composite unique constraints (`user_id` , `module_id` , `lesson_id`) to prevent duplicate record conflicts during rapid navigation.
- Expand API route handlers to support richer question types (multiple-choice, ordering).

Sushank Jung Rawal

- Research and prototype the 360° interactive panoramic viewer library (Photo Sphere Viewer / Three.js canvas) for the hazard recognition activity.
- Design wireframe layouts for the 360° scene HUD (Head-Up Display), inspection card modals, and hotspot pin markers.
- Prepare the visual slides and interface walkthrough for the client consultation.
- Author `03 System Requirements Analysis.pdf` (Task A1), drafting structured system flowcharts and data flow diagrams.

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4. Project Progress This Week

- **Core Full-Stack Prototype Functional:** The application boots cleanly, connects to live Supabase backend, authenticates users, and renders the interactive lesson wizard.
- **Requirements Baseline Established:** Task A1 drafts (User Requirements, PACT Analysis, System Architecture) completed and internally cross-referenced.
- **Project Schedule Operational:** Task A5 Gantt chart and individual plans for Rijan and Sushank structured and baseline dates established.
- **Corporate Brand Established:** Visual identity finalized around SARAS Primary Blue (`#0056D2`) and crane logo mark.

5. Issues Encountered and Planned Response

Issue Description	Severity	Impact	Planned Team Response
Progress Persistence Anomalies: Testing rapid navigation between lesson cards revealed duplicate progress entries and potential 409 conflict errors.	High	Risk of corrupted learner completion tracking and inaccurate dashboard status.	**Response:** Rijan will engineer a composite unique constraint in PostgreSQL (`ON CONFLICT (user_id, module_id, lesson_id) DO UPDATE`) in Migration 010.
Panoramic Library Performance: Preliminary inspection of heavy 3D WebGL engines indicated excessive bundle size and potential lag on low-end warehouse tablets.	Medium	Risk of poor responsiveness and battery drain on mobile devices.	**Response:** Sushank will test lightweight Photo Sphere Viewer bundles with compressed equirectangular imagery to ensure rapid browser rendering.

6. Formal Decisions and Action Items

Decisions

1. **D-04:** Adopt Photo Sphere Viewer for the 360° panoramic hazard inspection component due to its lightweight footprint and clean touch/keyboard event handling.
2. **D-05:** Structure the Week 3 client consultation strictly around five high-impact questions to respect Mr. Lama's commercial schedule.
3. **D-06:** Enforce automated TypeScript compilation checks on all GitHub pull requests.

Action Items

Action Item	Assigned Lead	Target Date	Status
Send formal Week 3 consultation invitation to Mr. Ninson Lama	Rijan Kapur Poudel	03 Aug 2026	Scheduled
Implement composite unique constraint on `user_progress`	Rijan Kapur Poudel	03 Aug 2026	Scheduled

Action Item	Assigned Lead	Target Date	Status
Prototype 360° panoramic canvas with sample equirectangular image	Sushank Jung Rawal	04 Aug 2026	In Progress
Complete structured system flowcharts for Task A1	Sushank Jung Rawal	05 Aug 2026	In Progress

Minutes submitted to Supervisory Tutor: Rupak Chaudhary | Recorded by Rijan Kapur Poudel | Approved by Sushank Jung Rawal